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EXP 1:-

```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char cmd[100];

    printf("Enter Linux command: ");
    scanf("%s", cmd);
    pid_t pid = fork();
    if(pid == 0)
    {
        printf("Child PID : %d\n", getpid());
        execlp(cmd, cmd, NULL);
        perror("Execution Failed");
    }
    else if(pid > 0)
    {
        printf("Parent PID : %d\n", getpid());
        wait(NULL);
        printf("Child process completed.\n");
    }
    else
    {
        printf("Fork Failed\n");
    }
    return 0;
}
```

OUTPUT:-

```
praneeth@PRANEETH:~$ nano 1.c
praneeth@PRANEETH:~$ gcc 1.c -o 1
praneeth@PRANEETH:~$ ./1
Enter Linux command: uname
Parent PID : 1129
Child PID : 1130
Linux
Child process completed.
praneeth@PRANEETH:~$ ./1
Enter Linux command: lscpu
Parent PID : 1140
Child PID : 1141
Architecture:                x86_64
  CPU op-mode(s):            32-bit, 64-bit
  Address sizes:             39 bits physical, 48 bits virtual
  Byte Order:                Little Endian
CPU(s):                      12
  On-line CPU(s) list:       0-11
Vendor ID:                   GenuineIntel
  Model name:                13th Gen Intel(R) Core(TM) i5-13420H
    CPU family:              6
    Model:                   186
    Thread(s) per core:      2
    Core(s) per socket:      6
    Socket(s):               1
    Stepping:                2
    BogoMIPS:                5222.40
    Flags:                   fpu vme de pse tsc msr pae mce cx8 apic sep mtr
                             r pge mca cmov pat pse36 clflush mmx fxsr sse s
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    Flags:                   fpu vme de pse tsc msr pae mce cx8 apic sep mtr
                             r pge mca cmov pat pse36 clflush mmx fxsr sse s
                             se2 ss ht syscall nx pdpe1gb rdtscp lm constant
                             _tsc rep_good nopl xtopology tsc_reliable nonst
                             op_tsc cpuid tsc_known_freq pni pclmulqdq vmx s
                             sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p
                             opcnt tsc_deadline_timer aes xsave avx f16c rdr
                             and hypervisor lahf_lm abm 3dnowprefetch ssbd i
                             brs ibpb stibp ibrs_enhanced tpr_shadow ept vpi
                             d ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi
                             2 erms invpcid rdseed adx smap clflushopt clwb
                             sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
                             vnmi umip waitpkg gfni vaes vpclmulqdq rdpid mo
                             vdiri movdir64b fsrm md_clear serialize ibt flu
                             sh_l1d arch_capabilities
Virtualization features:
  Virtualization:            VT-x
  Hypervisor vendor:         Microsoft
  Virtualization type:       full
Caches (sum of all):
  L1d:                       288 KiB (6 instances)
  L1i:                       192 KiB (6 instances)
  L2:                        7.5 MiB (6 instances)
  L3:                        12 MiB (1 instance)
NUMA:
  NUMA node(s):              1
  NUMA node0 CPU(s):         0-11
Vulnerabilities:
  Gather data sampling:      Not affected
  Ghostwrite:                Not affected
  Indirect target selection: Not affected
  Itlb multihit:             Not affected
  L1tf:                      Not affected
  Mds:                       Not affected
  Meltdown:                  Not affected
  Mmio stale data:           Not affected
  Old microcode:             Not affected
  Reg file data sampling:    Mitigation; Clear Register File
  Retbleed:                  Mitigation; Enhanced IBRS
  Spec rstack overflow:      Not affected
  Spec store bypass:         Mitigation; Speculative Store Bypass disabled v
                             ia prctl
  Spectre v1:                Mitigation; usercopy/swapgs barriers and __user
                              pointer sanitization
  Spectre v2:                Mitigation; Enhanced / Automatic IBRS; IBPB con
                             ditional; PBRSB-eIBRS SW sequence; BHI BHI_DIS_
                             S
```

```
                              S
  Srbds:                      Not affected
  Tsa:                        Not affected
  Tsx async abort:            Not affected
  Vmscape:                    Not affected
Child process completed.
praneeth@PRANEETH:~$ ./1
Enter Linux command: lsblk
Parent PID : 1151
Child PID : 1152
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda    8:0    0 356.9M  1 disk
sdb    8:16   0 159.4M  1 disk
sdc    8:32   0     2G  0 disk [SWAP]
sdd    8:48   0     1T  0 disk /mnt/wslg/distro
                                /
Child process completed.
praneeth@PRANEETH:~$ ./1
Enter Linux command: top
Parent PID : 1162
Child PID : 1163
top - 04:02:06 up 29 min,  1 user,  load average: 0.07, 0.04, 0.00
Tasks:  28 total,   1 running,  27 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.2 us,  0.2 sy,  0.0 ni, 99.6 id,  0.0 wa,  0.0 hi,  0.1 si,  0.0 st
MiB Mem :   7756.6 total,   6598.9 free,   1036.0 used,    275.4 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   6720.7 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
    283 mysql     20   0  983544 526140  44868 S   1.7   6.6   0:23.06 mysqld
   1163 praneeth  20   0    8284   6112   3944 R   0.3   0.1   0:00.05 top
      1 root      20   0   24252  15352  11540 S   0.0   0.2   0:01.25 systemd
      2 root      20   0    3180   2208   2072 S   0.0   0.0   0:00.04 init-systemd(Ub
      6 root      20   0    3180   2044   1940 S   0.0   0.0   0:00.00 init
     46 root      19  -1   50412  16940  15632 S   0.0   0.2   0:00.22 systemd-journal
     78 systemd+  20   0   22416  14396  11904 S   0.0   0.2   0:00.06 systemd-resolve
     87 root      20   0   35368  12328   9232 S   0.0   0.2   0:00.19 systemd-udevd
    162 root      20   0    2888   1952   1820 S   0.0   0.0   0:00.03 chronyd-starter
    163 root      20   0    4472   3116   2868 S   0.0   0.0   0:00.00 cron
    164 message+  20   0    8792   5444   4716 S   0.0   0.1   0:00.09 dbus-daemon
    172 root      20   0   44896  29868  14728 S   0.0   0.4   0:00.21 networkd-dispat
    180 root      20   0   18436   9396   8152 S   0.0   0.1   0:00.10 systemd-logind
    193 syslog    20   0  220548   5920   4628 S   0.0   0.1   0:00.11 rsyslogd
    219 root      20   0    5492   2720   2536 S   0.0   0.0   0:00.01 agetty
    247 _chrony   20   0   20424   7812   6212 S   0.0   0.1   0:00.07 chronyd
    265 root      20   0  123460  32704  18088 S   0.0   0.4   0:00.17 unattended-upgr
    267 _chrony   20   0   12096   2428   1788 S   0.0   0.0   0:00.00 chronyd
    474 root      20   0    3184   1112    980 S   0.0   0.0   0:00.00 SessionLeader
    475 root      20   0    3200   1248   1108 S   0.0   0.0   0:00.18 Relay(480)
    480 praneeth  20   0    6268   5536   3864 S   0.0   0.1   0:00.13 bash
    481 root      20   0    8648   5532   4688 S   0.0   0.1   0:00.00 login
    526 praneeth  20   0   22504  13392  10956 S   0.0   0.2   0:00.17 systemd
    528 praneeth  20   0   22912   4120   2120 S   0.0   0.1   0:00.00 (sd-pam)
    558 praneeth  20   0    6136   5416   3864 S   0.0   0.1   0:00.05 bash
```

```
    901 root      20   0 1792300  12100   9444 S   0.0   0.2   0:00.14 wsl-pro-service
   1017 polkitd   20   0  306444   8440   7624 S   0.0   0.1   0:00.13 polkitd
   1162 praneeth  20   0    2768   1880   1784 S   0.0   0.0   0:00.00 1
   1163 praneeth  20   0    8284   6112   3944 R   0.0   0.1   0:00.05 top
```

EXP 2:-

```c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
        int src, dest;
        char buffer[1024];
        int bytes;

        src = open("input.txt", O_RDONLY);
        if(src < 0)
{
        printf("Cannot open source file\n");
         return 1;
}

dest = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
while((bytes = read(src, buffer, sizeof(buffer))) > 0)
{
        write(dest, buffer, bytes);
}
 close(src);
 close(dest);
  printf("File copied successfully.\n");
 return 0;
}
```

OUTPUT:-

```
praneeth@PRANEETH:~$ nano 2.c
praneeth@PRANEETH:~$ gcc 2.c -o 2
praneeth@PRANEETH:~$ ./2
Cannot open source file
praneeth@PRANEETH:~$
```

EXP 3:-

```c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid = fork();

    if(pid == 0)
    {
        printf("Child Process\n");
        printf("PID  : %d\n", getpid());
        printf("PPID : %d\n", getppid());

        sleep(10);
    }
    else
    {
        printf("Parent Process\n");
        printf("PID  : %d\n", getpid());

        wait(NULL);
    }

    return 0;
}
```

OUTPUT:-

```
praneeth@PRANEETH:/mnt/c/Users/praneeth$ gcc 3.c -o 3
praneeth@PRANEETH:/mnt/c/Users/praneeth$ ./3
Parent Process
PID  : 607
Child Process
PID  : 608
PPID : 607
```

EXP 4:-

```c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int i;

    for(i=0;i<3;i++)
    {
        if(fork()==0)
        {
            printf("Child %d PID=%d\n",i+1,getpid());
            return 0;
        }
    }

    for(i=0;i<3;i++)
    {
        wait(NULL);
    }

    printf("All children completed.\n");

    return 0;
}
```

OUTPUT:-

```
praneeth@PRANEETH:/mnt/c/Users/praneeth$ nano 4
praneeth@PRANEETH:/mnt/c/Users/praneeth$ gcc 4.c -o 4
praneeth@PRANEETH:/mnt/c/Users/praneeth$ ./4
Child 1 PID=691
Child 2 PID=692
Child 3 PID=693
All children completed.
praneeth@PRANEETH:/mnt/c/Users/praneeth$
```